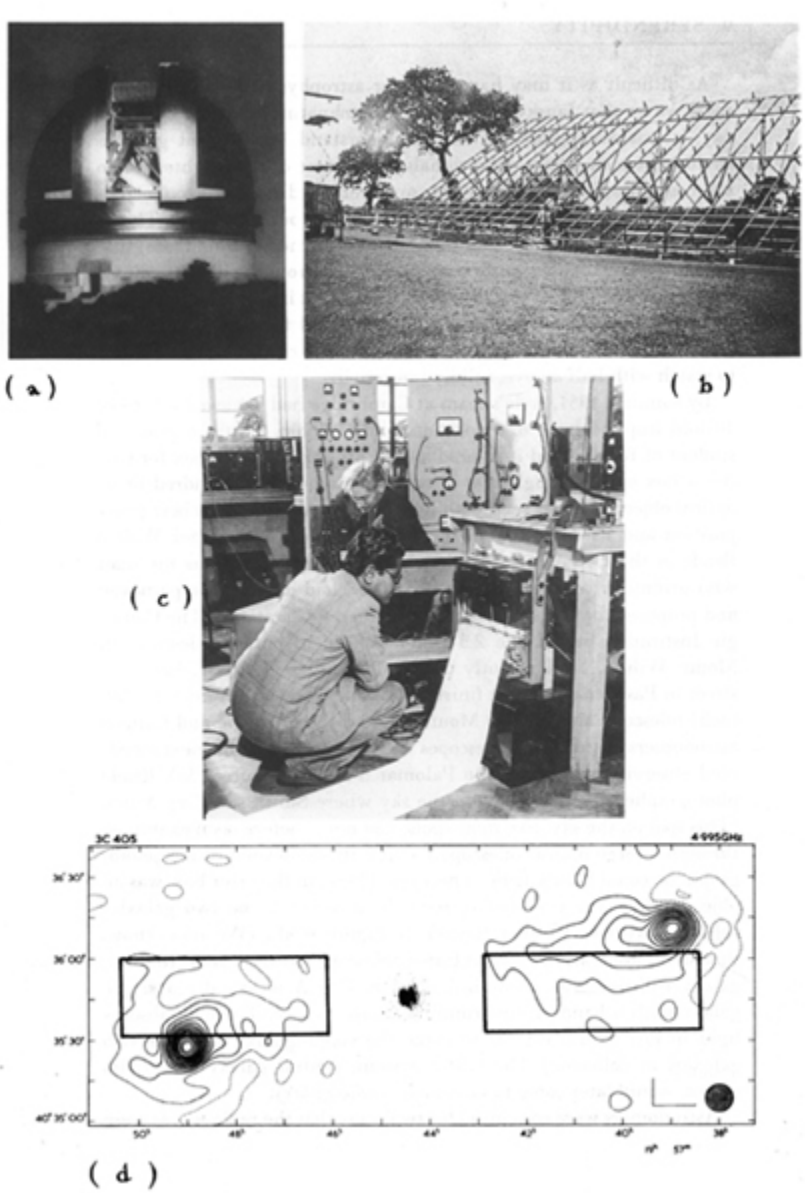
contours confirm the 1953 conclusion that the radio waves come from gigantic lobes of gas on either side of the “colliding galaxies.” How both of these enormous lobes can be powered by a single, gigantic black hole will become a major issue later in this chapter.



9.3 The discovery that Cyg A is a distant *radio galaxy:* (a) The 5-meter optical telescope used in 1951 by Baade to discover that Cyg A is connected with what appeared to be two colliding galaxies. (b) The radio interferometer at Jodrell Bank used in 1953 by Jennison and Das Gupta to show that the radio waves are coming from two giant lobes outside the colliding galaxies. The interferometer’s two antennas (each an array of wires on a wooden framework) are shown here side by side. In the measurements, one was put on a truck and moved around the countryside, while the other remained behind, at rest on the ground. (c) Jennison and Das Gupta, inspecting the radio data in the control room of their interferometer. (d) The two giant lobes of radio